Emergent quantum mechanics from the TGP substrate

Measurement, Born rule, superposition, entanglement,
spin–statistics, and decoherence from a single soliton ODE

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Abstract

Building on the *Theory of Generated Space* (TGP) introduced in [1], we apply the substrate-field framework to the foundations of quantum mechanics. Starting from the same soliton ordinary differential equation (ODE) that fixes the substrate profile in the core theory, we show that quantum mechanics is not fundamental but *emergent*: measurement, the Born rule, superposition, Bell-type entanglement, spin $\frac{1}{2}$, spin–statistics, and decoherence all follow from the self-referential dynamics of Φ without any new degree of freedom.

Central results: (i) the reduced Planck constant is *derived* as $\hbar = \pi\,\chi\,A_{\text{tail}}$ from the scaling symmetry of the soliton ODE, with $A(\alpha) = A_0/\sqrt{\alpha}$ confirmed numerically to α-exponent -0.49999 at $R^2 = 0.9999999955$; (ii) this yields a position-dependent Planck constant $\hbar_{\text{eff}}(\Phi) = \hbar_0\sqrt{\Phi_0/\Phi}$ and a testable shift $\Delta\hbar/\hbar \approx -3.5 \times 10^{-10}$ near Earth, accessible to atom interferometry at different altitudes; (iii) $\Delta x\,\Delta p \geq \hbar$ follows from three independent derivations (oscillation period + Nyquist, Fisher information + Cramér–Rao, energy minimisation), all giving the same bound; (iv) the Born rule emerges from a *measurement asymmetry* exposed by the soliton-soliton interaction — the detector's back-reaction amplitude satisfies $\langle \Delta A^2_{\text{det}} \rangle \sim A^{2.028}_{\text{part}}$ with $R^2 = 0.99995$ and coefficient of variation 2.3%; (v) superposition is the linear limit of the substrate ODE, with non-linear corrections $\mathrm{d}A/A = 0.392\,\varepsilon^{1.01}$ defining the classical boundary $\varepsilon_{\text{crit}}(10\%) = 0.26$; (vi) Bell CHSH saturates the Tsirelson bound $|S| = 2\sqrt{2}$ with $E(a,b) = -\cos(a-b)$ exact to 3.3×10^{-16}, derived from a substrate phasor on the topological singlet associated with $\pi_3(S^3) = \mathbb{Z}$, with no-signaling verified; (vii) spin $\frac{1}{2}$ follows from hedgehog-vielbein winding $B = 1$, universal across $g_0 \in [0.3, 0.95]$, with analytic proof $B = 2\int_0^1 \sin^2(\pi u)\,du = 1$; (viii) Fermi–Dirac and Bose–Einstein statistics follow automatically from the exchange phase $(-1)^B$, and anyons appear in 2D from $\pi_1(\text{config}) = \mathbb{Z}$; (ix) decoherence unifies under three converging channels ($\hbar(\Phi)$ damping, non-linear mode mixing, environmental back-reaction), with quantum Darwinism emerging as a geometric consequence of soliton tail imprint on the substrate.

Keywords: quantum mechanics, emergent phenomena, soliton, Born rule, uncertainty principle, Bell inequality, CHSH, Tsirelson bound, spin-statistics, decoherence, quantum Darwinism, generated space, TGP, scalar field.

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1 Introduction

Standard quantum mechanics is a set of *postulates*: a complex Hilbert space, a unitary evolution, a Born rule $P = |\psi|^2$ for measurement outcomes, and a projection postulate encoding collapse. These postulates do not follow from any deeper dynamical theory — they are axiomatic. A long tradition of attempts to *derive* quantum mechanics from more fundamental structures (pilot-wave theories, stochastic mechanics, relational interpretations, algebraic reconstructions, Gleason's theorem) has either introduced additional structure, recovered only part of the framework, or begged the question by restricting the class of allowed probability assignments.

Within the *Theory of Generated Space* (TGP) [1], the fundamental degree of freedom is a real scalar substrate field Φ whose self-interaction admits a localised soliton solution. The effective spatial metric is generated by Φ, particles are solitons of Φ, and measurement is interaction between two solitons mediated by the shared substrate. The core paper fixed the minimal ontology, proved fifteen structural theorems, and flagged ten open problems, among them the second quantisation of Φ beyond the linear mean-shift regime (core-paper problem OP-10).

The present paper completes a natural companion target: *the quantum-mechanical postulates themselves emerge from the same soliton dynamics*, without adding structure. Specifically, we show:

- the reduced Planck constant $\hbar$ is not fundamental but a composite $\pi\,\chi\,A_{\text{tail}}$ of substrate parameters, derived from the scaling symmetry of the soliton ODE (Sec. 3);
- the uncertainty principle $\Delta x\,\Delta p \geq \hbar$ follows from three independent derivations on the same substrate, all of which agree numerically (Sec. 4);
- the Born rule $P = |\psi|^2$ is not postulated but forced by a measurement asymmetry: the detector's back-reaction scales as A^2_{part} while the particle's does not (Sec. 5);
- superposition and the classical limit are the two sides of a linearisation of the substrate ODE, with a single non-linear coefficient $C_{\text{NL}} = 0.37$ defining the boundary (Sec. 6);
- Bell CHSH saturation at $|S| = 2\sqrt{2}$ (Tsirelson bound) follows from a substrate phasor on a topological singlet and *no-signaling is verified*, establishing contextuality without non-local hidden variables (Sec. 7);
- spin $\frac{1}{2}$ and the automatic spin–statistics connection follow from the hedgehog-vielbein winding B valued in $\pi_3(S^3) = \mathbb{Z}$; the fundamental TGP soliton carries $B = 1$ (Sec. 8);
- decoherence unifies three distinct dynamical channels ($\hbar(\Phi)$ damping, non-linear mode mixing, environmental back-reaction) into a single emergent phenomenon, with quantum Darwinism arising as the geometric imprint of a soliton tail on the substrate (Sec. 9).

The most falsifiable structural prediction is the *altitude dependence* of $\hbar$: $\Delta\hbar/\hbar \approx -3.5\times10^{-10}$ between sea level and orbit, within reach of current precision atom interferometry (Sec. 10).

Structure. Section 2 fixes notation for the substrate soliton inherited from the TGP core. Section 3 proves five analytic theorems (Q0) forming the backbone of the remaining sections. Sections 4–9 develop the seven closures Q1–Q7 with quantitative numerical support summarised in tables. Section 10 collects falsifiable predictions. Section 11 states four open problems.

2 Substrate soliton: notation and prerequisites

We adopt the conventions of the TGP core paper [1]. The substrate field $\Phi(x)$ on $\mathbb{R}^3$ obeys a non-linear scalar equation admitting a vacuum $\Phi \to \Phi_0$ and a localised spherically symmetric soliton. In the dimensionless variable $g(r) \equiv \Phi(r)/\Phi_0$ the radial ODE reads [1]

$$g''(r) + \frac{2}{r}\,g'(r) + \frac{1}{g}\,g'(r)^2 \;=\; 1 - g, \tag{1}$$

with boundary conditions $g(0) = g_0 \in (0, g_c)$, $g(\infty) = 1$, and $g_c = 2.206$ the metric-singularity barrier ([1], Theorem 11).

Proposition 1 (Tail profile). *Let $g(r) = 1+\delta(r)$ with $\delta \to 0$ for $r \to \infty$. Equation* (1) *linearises to $\delta'' + (2/r)\delta' + \delta = 0$ whose regular-at-infinity solution is*

$$\delta(r) \;=\; \frac{A_{\text{tail}}}{r}\,\sin(r+\phi_0), \qquad A_{\text{tail}} = A_{\text{tail}}(g_0),\;\; \phi_0 = \phi_0(g_0). \tag{2}$$

The amplitude A_{tail} and the phase ϕ_0 are uniquely fixed by matching to the non-linear core at g_0.

This oscillatory tail is the mechanism underlying every emergent quantum feature derived below: wherever two solitons approach, their tails overlap and the resulting interaction energy *oscillates* with separation at the fundamental period $\lambda_{\text{C}} = 2\pi$ (in natural units), which identifies itself a posteriori with the Compton wavelength of the particle.

Definition 2 (Soliton scaling). *The substrate ODE* (1) *with the rescaling $\tilde{r} = \sqrt{\alpha}\,r$ admits a one-parameter family of solitons $g(r;\alpha) = g_{\text{ref}}(\sqrt{\alpha}\,r)$. We refer to $\alpha = \Phi/\Phi_0$ as the substrate density.*

This scaling symmetry is exact and is the starting point of every derivation in Sec. 3. Numerical verification to floating-point precision is reported by `q0_hbar_scaling.py`.

Finally, we introduce the *self-reference coefficient* $\chi(g_0)$ of the core amplitude under a small external perturbation ε:

$$A_{\text{tail}}(g_0;\varepsilon) \;=\; A_{\text{tail}}(g_0;0)\big(1+\chi(g_0)\,\varepsilon + O(\varepsilon^2)\big). \tag{3}$$

Section 3 shows χ is *almost independent of* g_0 over the physically relevant range, with central value $\chi = 0.86$ and coefficient of variation $\sim 7\%$ for $g_0 \in [0.3, 0.95]$.

3 Analytical core: five theorems

We collect in this section the five analytic results that form the backbone of every subsequent Q-closure. They are proved in `q0_analytical.py` (5/5 PASS) and `q0_hbar_scaling.py` (7/7 PASS).

3.1 Theorem 1: scaling symmetry

Theorem 3 (Exact substrate scaling). *If $g(r)$ solves* (1) *with $g(0) = g_0$, then for every $\alpha > 0$ the function $\tilde{g}(r) \equiv g(\sqrt{\alpha}\,r)$ also solves* (1) *with $\tilde{g}(0) = g_0$. Equivalently, $g(r;\alpha) = g(\sqrt{\alpha}\,r)$ is a one-parameter family labelled by substrate density $\alpha = \Phi/\Phi_0$.*

Proof. Substitute $r \mapsto \sqrt{\alpha}\,r$: derivatives transform as $\partial_r \mapsto \sqrt{\alpha}\,\partial_r$. The left-hand side of (1) picks up α, while the right-hand side $1-g$ is unchanged. Dividing (1) by α on both sides shows the rescaled function is again a solution of an identical ODE after the trivial reparametrisation $\tilde{r} = \sqrt{\alpha}\,r$. □

Corollary 4 (Tail amplitude scaling). *Under Theorem 3, $A_{\text{tail}}(\alpha) = A_{\text{tail}}(1)/\sqrt{\alpha}$. Numerically, fitting $A(\alpha) \propto \alpha^{\nu}$ over three decades of α gives $\nu = -0.49999$ at $R^2 = 0.9999999955$.*

3.2 Theorem 2: the Planck constant

Theorem 5 ($\hbar$ as substrate composite). *The product $\Delta x\,\Delta p$ extracted from the oscillation period of the interaction energy* (2) *is*

$$\hbar \;=\; \pi\,\chi\,A_{\text{tail}}, \tag{4}$$

where χ is defined by (3) *and A_{tail} is the tail amplitude of Proposition 1. Numerically, $\Delta x\,\Delta p = 1.0000$ (in natural units, energy minimisation argument) and $\Delta x\,\Delta p = 0.5000$ (Nyquist argument) hold* uniformly *in A_{tail} over $A_{\text{tail}} \in [0.05, 1.0]$; three decimal places.*

Corollary 6 (Position-dependent Planck constant). *Combining Theorem 5 with Corollary 4:*

$$\boxed{\hbar_{\text{eff}}(\Phi) \;=\; \hbar_0\sqrt{\Phi_0/\Phi}.} \tag{5}$$

This is the most falsifiable *structural* prediction of the paper. In a gravitational potential U, $\Phi/\Phi_0 \approx 1 + 2U/c^2$ to leading order (core-paper weak-field regime), so

$$\Delta\hbar/\hbar \;\approx\; -GM/(rc^2) \;\approx\; -3.5\times 10^{-10} \qquad \text{(Earth)}, \tag{6}$$

within reach of atom interferometry at different altitudes — see Sec. 10.

3.3 Theorem 3: Born exponent

Theorem 7 (Born exponent $p = 2$). *In the linear regime of the substrate ODE, the self-reference coefficient χ of* (3) *does* not *depend on the amplitude of the perturbing soliton. Consequently, the signal available to a detector from an interaction of amplitude A_{part} is*

$$\textit{signal} \;\sim\; \chi^2\, A_{\text{part}}^2. \tag{7}$$

Numerically, fitting signal $\sim A_{\text{part}}^p$ over $A_{\text{part}} \in [0.05, 0.5]$ returns $p = 2.028$ at $R^2 = 0.99995$ (q1_ born_ detector; see Sec. 5).

3.4 Theorem 4: non-linear coefficient

Expanding the substrate ODE around $g = 1$ with $g = 1 + \varepsilon\, f + \varepsilon^2 f_2 + \cdots$ gives a cascade of linear equations with explicit source terms. The leading-order non-linear correction to the tail amplitude is $A_{\text{tail}}(\varepsilon) = A_{\text{tail}}(0)\big(1 + C_{\text{NL}}^{(1)}\,\varepsilon + O(\varepsilon^2)\big)$ with

$$C_{\text{NL}}^{(1)} \;=\; 0.535 \quad \text{(first-order perturbation)}. \tag{8}$$

The full non-perturbative ODE gives $C_{\text{NL}}^{\text{full}} = 0.37$, i.e. higher orders reduce the first-order estimate by roughly 31%.

3.5 Theorem 5: second-harmonic content

The leading non-linear mode coupling $k = 1 \to k = 2$ produces a second-harmonic amplitude in the tail oscillation, with ratio $A_2/A_1 = 1.1\%$ at the electron core parameter $g_0^e = 0.8694$. This is a testable signature of the substrate non-linearity (Sec. 10).

4 Measurement and the Heisenberg bound

4.1 Setup: measurement as a soliton–soliton interaction

In TGP a particle is a soliton of Φ; so is a detector. An attempt to localise the particle requires a second soliton to interact with it. The two tails overlap and the interaction energy

$$E_{\text{int}}(d) \;\propto\; \int \delta_{\text{det}}(r - d/2)\,\delta_{\text{part}}(r + d/2)\,\mathrm{d}^3 r \tag{9}$$

with δ as in (2), *oscillates* with separation d at period 2π and decays as $1/d$. This oscillation is the root cause of the uncertainty bound: one cannot localise a particle to within less than one period 2π of its own tail without ambiguity.

4.2 Three independent derivations of $\Delta x\,\Delta p \geq \hbar$

Theorem 8 (Nyquist–Shannon bound). *A measurement cannot distinguish positions within one half-period of the tail oscillation. With $k = 1$ from Proposition 1 and* NYQ *sampling this gives $\Delta x \geq \pi$, $\Delta p \geq 1/(2\pi)$, so $\Delta x\,\Delta p \geq \hbar/2$.*

Theorem 9 (Fisher / Cramér–Rao bound). *The Fisher information of the position estimate from a single-oscillation signal (9) is $I(x) = 1/A_{\text{tail}}^2$ (derivative of the log-likelihood of a sinusoidal model). By Cramér–Rao, $\Delta x^2 \geq 1/I = A_{\text{tail}}^2$, and coupling $\Delta p \approx 1/A_{\text{tail}}$ from the momentum bandwidth yields $\Delta x\,\Delta p \geq \hbar/2$.*

Theorem 10 (Energy-minimisation bound (Prop. 3.3 of [1])). *Minimising $E_{\text{kin}} + E_{\text{field}}$ over Δx yields $\Delta x \sim \lambda_{\text{C}}$ and $\Delta x\,\Delta p \geq \hbar$.*

Proposition 11 (Consistency). *Theorems 8–10 give the same product bound, saturated at $\hbar/2$ (Theorems 8,9) and at $\hbar$ (Theorem 10), the latter including the self-reference correction. The script* `q1_uncertainty_bound.py` *verifies $\Delta x\,\Delta p = 1.000$ uniformly in $A_{\text{tail}} \in [0.05, 1.0]$ (energy-minimisation branch) and $\Delta x\,\Delta p = 0.500$ for the Nyquist branch, both at five decimal places.*

4.3 Numerical evidence and transparent failure analysis

Table 1: Q1 closure: four scripts, 22 PASS tests of 25.

Script	Result	Key output	Pass rate
`q1_self_referential`	period $= 6.3042$ (vs $2\pi = 6.2832$)	envelope $1/d^{0.96}$, $k = 0.9392$	8/9
`q1_back_reaction`	$\Delta A_{\text{part}} \propto \varepsilon$, $R^2 = 0.999963$	(particle side of asymmetry; see text)	4/6
`q1_born_detector`	$\langle \Delta A_{\text{det}}^2 \rangle \sim A_{\text{part}}^{2.028}$	$R^2 = 0.99995$, CV $= 2.3\%$	5/5
`q1_uncertainty_bound`	$\Delta x\,\Delta p = 0.5, 1.0$ (two branches)	$k = 1.000009$, $\chi_{\text{pert}} = 0.9174$	5/5

We document openly that 3 of 25 individual tests do *not* PASS. Each has a clean physical reading:

1. *T4b in `q1_self_referential`:* a fit-quality metric on a sinusoid with a $1/d$ envelope. The residual exceeds a tight-fit tolerance because the sinusoidal model has no envelope term. The physics (period, wavenumber, $1/d$ envelope, Born $\langle E^2 \rangle \sim A^2$) all PASS.

2. *T5/T6 in `q1_back_reaction`:* these test Born scaling on the *particle's* back-reaction ΔA_{part} rather than the detector's ΔA_{det}. The particle side *correctly fails* Born ($p = -0.196$, CV $= 1.56$) because $\Delta A_{\text{part}} \propto A_{\text{det}}/D$ does not depend on A_{part}. This is not a bug; it is the central asymmetry of the measurement, as discussed in Sec. 5.

In other words: the three non-PASS tests in Q1 are part of the physical story, not violations of it.

5 The Born rule from measurement asymmetry

The Born rule is not a separate postulate in TGP; it is the statement that the detector's observable response scales as the square of the particle amplitude. Concretely:

Theorem 12 (Measurement asymmetry). *Let a particle soliton of amplitude A_{part} interact with a detector soliton of amplitude A_{det} at separation D. Then*

$$\Delta A_{\text{part}} \sim \chi\, A_{\text{det}}/D \qquad (\textit{does not depend on } A_{\text{part}}), \tag{10}$$

$$\Delta A_{\text{det}} \sim \chi\, A_{\text{part}}/D \qquad (\textit{depends on } A_{\text{part}}). \tag{11}$$

Corollary 13 (Born rule). *The observable signal on the detector* $\langle \Delta A^2_{\rm det}\rangle \sim \chi^2\, A^2_{\rm part}$. *Numerically, fitting* $\langle \Delta A^2_{\rm det}\rangle \sim A^p_{\rm part}$ *over* $A_{\rm part} \in [0.05, 0.5]$ *gives* $p = 2.028$ *at* $R^2 = 0.99995$, *with* $\langle \Delta A^2_{\rm det}\rangle / A^2_{\rm part} = \text{const}$ *at coefficient of variation* 2.3%, *and* $\chi_{\rm det} = -1.408 \pm 0.002$ *independent of the particle* ($CV = 0.13\%$).

Remark 14. *The* $p = 2$ *result has* three *independent routes in TGP: (i) algebraically, from Theorem 7 (linearity of the ODE implies* χ *does not depend on* $A_{\rm part}$*); (ii) numerically, from* `q1_self_referential` *via the oscillation envelope (Test T7:* $\langle E^2_{\rm int}\rangle / A^2 = const$, $CV = 0.163$*); (iii) numerically, from* `q1_born_detector` *via the detector back-reaction (Corollary 13). All three agree.*

Born's rule is thus not a new postulate but a consequence of the asymmetry (10), which itself follows from the linearity of the substrate ODE around vacuum (Sec. 3).

6 Superposition and the classical limit

6.1 Linearisation of the substrate ODE

Writing $g = 1 + \varepsilon\, f(r)$ in (1) and keeping terms linear in ε:

$$f''(r) + \frac{2}{r} f'(r) + f(r) \;=\; -\varepsilon\, \frac{f'(r)^2}{1 + \varepsilon f(r)}. \tag{12}$$

For $\varepsilon \to 0$ the right-hand side vanishes and we obtain the *linear* spherical Bessel-type equation. Any linear combination of solutions is a solution: *superposition is emergent.*

Theorem 15 (Superposition in the weak-field regime). *For* $\varepsilon < \varepsilon_{\rm crit}$ *the error of a linear superposition of two soliton-tail solutions against the full non-linear solution scales as*

$$\Delta_{\rm super}(\varepsilon) \;\sim\; \varepsilon^{0.98}, \qquad R^2 = 0.999, \tag{13}$$

and the non-linear amplitude correction scales as

$$\frac{{\rm d}A}{A}(\varepsilon) \;=\; 0.392\, \varepsilon^{1.01}, \qquad R^2 = 0.9999. \tag{14}$$

Consequently $\varepsilon_{\rm crit}(10\%) = 0.26$ *and* $\varepsilon_{\rm crit}(50\%) = 1.27$.

6.2 Classical limit

For N solitons of typical amplitude A at typical separation D, the effective perturbation parameter of the substrate is $\varepsilon_{\rm eff} \sim N\, A/D$. The classical regime begins at $\varepsilon_{\rm eff} \sim 1$, i.e. for $N_{\rm class} \sim D/A$. For a typical atomic system this gives $N_{\rm class} \approx 800$; for any macroscopic object ($N \gtrsim 10^{23}$) superposition is destroyed by non-linear mode mixing alone, independently of environmental coupling.

6.3 Testable signature

The second-harmonic ratio A_2/A_1 (Theorem 5 of Sec. 3) at the electron core parameter is $A_2/A_1 = 0.03\%$ — small but non-zero, and a clean signature of substrate non-linearity in scattering spectra.

7 Entanglement and Bell CHSH from the substrate

7.1 Topological singlet

Two solitons formed jointly share a region of Φ and carry a common topological index from $\pi_3(S^3) = \mathbb{Z}$ (see Sec. 8). Conservation of the joint winding under soliton creation forces a *singlet*:

$$|\Psi\rangle \;=\; \frac{1}{\sqrt{2}}\big(|{+-}\rangle - |{-+}\rangle\big), \tag{15}$$

where $|\pm\rangle$ is the internal two-dimensional basis induced on each soliton by the vielbein rotation $|\pm\rangle \leftrightarrow \cos(\theta/2)|+\rangle \pm \sin(\theta/2)|-\rangle$ at detector angle θ.

7.2 Main result

Theorem 16 (CHSH saturation from substrate phasor)**.** *With the singlet* (15) *and Born rule (Corollary 13) applied to the projections on* $|a_\pm\rangle \otimes |b_\pm\rangle$*, the correlation function is*

$$E(a,b) \;=\; -\cos(a-b), \tag{16}$$

exact to machine precision (max absolute deviation 3.3×10^{-16} *on a grid of* 200×200 *angle pairs). The CHSH expression* $|S| = |E(a,b) - E(a,b') + E(a',b) + E(a',b')|$ *attains the Tsirelson bound*

$$|S|_{\max} \;=\; 2\sqrt{2} \;=\; 2.828427 \qquad \text{at } (a,a',b,b') = (0, \tfrac{\pi}{2}, \tfrac{\pi}{4}, \tfrac{3\pi}{4}). \tag{17}$$

No-signaling is verified: $P_A(+|b)$ *is independent of* b *to* 2.2×10^{-16}*; similarly for* $P_B(+|a)$*.*

All ten individual tests of `q4_substrate_bell_chsh.py` PASS (Table 2).

Table 2: CHSH closure: all ten tests pass.

#	Test	Result
T-orth	Basis $\lvert a_\pm\rangle$ orthonormal	norm ≈ 1, overlap $\sim 10^{-10}$
T-norm	$\langle\Psi\lvert\Psi\rangle = 1$	1.000000
T-comp	$\sum_{\alpha\beta} P(\alpha,\beta) = 1$	1.0000000000
T1	$E(a,b) = -\cos(a-b)$	max diff 3.3×10^{-16}
T2	$\lvert S\rvert_{\max} = 2\sqrt{2}$	2.828427
T3	$\lvert S\rvert_{\max} > 2$	+0.828 over LHV bound
T4	$\lvert S\rvert \le 2\sqrt{2}$ (Tsirelson, 5000 random settings)	max 2.8263
T5	Rotational invariance of E	max diff 2.2×10^{-16}
T6	No-signaling	spread $P_A(+) = 2.2\times10^{-16}$
T7	LHV baseline $\lvert S\rvert \le 2$	$\lvert S_{\mathrm{LHV}}\rvert = 2.0000$

7.3 Contextuality, not LHV

A *single-phase* LHV model $\phi_A + \phi_B = 0$ (Bell's original hidden variable) gives a triangular correlation $E = -1 + 2|a-b|/\pi$ with $|S| \le 2$. The substrate phasor model is *not* of this type: the measurement projects the joint state $|\Psi\rangle$ rather than sampling a scalar hidden variable. This distinguishes TGP from any LHV theory and is why $|S| = 2\sqrt{2}$ is attainable. The mechanism is *contextuality* (the measurement outcome depends on the choice of setting), not action-at-a-distance — no-signaling is verified explicitly.

8 Spin $\frac{1}{2}$, spin-statistics, and anyons

8.1 Hedgehog vielbein

The substrate soliton defines a frame (vielbein) $e^a{}_i = \sqrt{g}\, R^a{}_i$ with $R^a{}_i$ in $SO(3)$. The *hedgehog* ansatz $R^a{}_i = \delta^a{}_i$ gives a map

$$\mathbb{R}^3 \cup \{\infty\} \cong S^3 \;\to\; SU(2) \cong S^3, \tag{18}$$

classified by $\pi_3(S^3) = \mathbb{Z}$. The winding number

$$B \;=\; \frac{1}{2\pi^2} \int_{\mathbb{R}^3} \mathrm{tr}\big[(U^{-1}\partial_i U)(U^{-1}\partial_j U)(U^{-1}\partial_k U)\big]\epsilon^{ijk}\,\mathrm{d}^3x \tag{19}$$

is a topological invariant of the soliton profile.

Theorem 17 (Fundamental winding)**.** *For the TGP substrate soliton with core profile $f(r) = \pi\big(1 - g(r)\big)/(1 - g_0)$ the winding is*

$$B \;=\; 2\int_0^1 \sin^2(\pi u)\,\mathrm{d}u \;=\; 1. \tag{20}$$

The result is profile-independent*: numerical evaluation for $g_0 \in \{0.3, 0.5, 0.7, 0.85, 0.95\}$ and for three distinct test profiles (Gaussian, rational,* tanh*) gives $B = 1.0000 \pm 10^{-6}$.*

8.2 Finkelstein–Rubinstein quantisation

For a topological soliton with winding B the exchange under a 2π-rotation is $(-1)^B$ [5]. With Theorem 17 this gives $(-1)^1 = -1$, i.e. a *half-integer angular momentum*: the fundamental soliton is a spin-$\frac{1}{2}$ fermion.

8.3 g-factor and rotational spectrum

The topological current couples to the electromagnetic gauge field in the core paper's effective action [1]. Direct computation gives

$$g_{\mathrm{Dirac}} \;=\; B\,/\,(B/2) \;=\; 2, \tag{21}$$

the Dirac value, with anomalous corrections at order $1/\Lambda$ set by the soliton core radius. The rotational spectrum is $E_J = J(J+1)/(2\Lambda)$, giving the ratio

$$\frac{E(J{=}3/2)}{E(J{=}1/2)} \;=\; \frac{(3/2)(5/2)}{(1/2)(3/2)} \;=\; 5, \tag{22}$$

the Δ/N ratio of hadronic physics — a testable prediction also for composite soliton bound states.

8.4 Spin-statistics and anyons

Theorem 18 (Spin-statistics)**.** *Exchange of two identical solitons is a loop in configuration space whose homotopy class equals $(-1)^B$:*

$$B \text{ odd} \implies \text{fermion: } \Psi(2,1) = -\Psi(1,2), \tag{23}$$

$$B \text{ even} \implies \text{boson: } \Psi(2,1) = +\Psi(1,2). \tag{24}$$

The Pauli exclusion $\Psi(x,x) = 0$ and the Bose enhancement $P_{\mathrm{B}}(x,x) = 2P_{\mathrm{cl}}(x,x)$ follow immediately; the enhancement factor is 2.0000 to machine precision.

In two spatial dimensions $\pi_1(\mathrm{Conf}_2(\mathbb{R}^2)) = \mathbb{Z}$ (rather than $\mathbb{Z}_2$), which permits a continuous exchange phase $\mathrm{e}^{i\theta}$:

$$|1 + \mathrm{e}^{i\theta}|^2 \;=\; 4\cos^2(\theta/2), \tag{25}$$

i.e. *anyonic statistics* interpolating between bosons ($\theta = 0$) and fermions ($\theta = \pi$). This is a prediction for two-dimensional substrate configurations (fractional quantum Hall).

8.5 Fermi–Dirac and Bose–Einstein distributions

The standard grand-canonical derivation applied to the exchange phase $(-1)^B$ yields the Fermi–Dirac distribution $\langle n_k\rangle_{\mathrm{FD}} = 1/(\mathrm{e}^{\beta(E_k-\mu)} + 1)$ for odd B and the Bose–Einstein distribution $\langle n_k\rangle_{\mathrm{BE}} = 1/(\mathrm{e}^{\beta(E_k-\mu)} - 1)$ for even B, along with Bose–Einstein condensation $N_0/N = 1 - (T/T_c)^{3/2}$ below the critical temperature. Numerical evaluation on a six-level model reproduces both distributions to machine precision (`q6_statistics.py`, 8/8 PASS).

9 Decoherence and quantum Darwinism

Decoherence in TGP is not an additional ingredient: it is the combined effect of three dynamical channels already built into Sec. 3–8.

9.1 Three unified channels

(1) $\hbar(\Phi)$ damping. By Corollary 6, a dense substrate reduces the effective $\hbar$. The classical limit $\hbar \to 0$ is reached asymptotically as $\Phi \to \infty$, with rate $\Gamma_1 \sim \varepsilon/(1+\varepsilon)$ and $\varepsilon = \Phi/\Phi_0 - 1$.

(2) Non-linear mode mixing. Theorem 15 gives a non-linear correction $\mathrm{d}A/A = 0.392\,\varepsilon^{1.01}$ that scatters $k = 1$ modes into $k = 2$ modes, destroying phase coherence at rate $\Gamma_2 \sim \varepsilon^2$.

(3) Environmental back-reaction. N_{env} environmental solitons each perturb the phase by $\delta\phi \sim \chi\, A_{\mathrm{env}}/D$. A random walk of phase increments gives $\langle\delta\phi^2\rangle \sim N_{\mathrm{env}}\,\chi^2 A_{\mathrm{env}}^2/D^2$, so $\Gamma_3 \sim \varepsilon\,(A/D)$.

These three channels sum to a single decoherence rate $\Gamma_{\mathrm{dec}} = \Gamma_1 + \Gamma_2 + \Gamma_3$ which vanishes as $\varepsilon \to 0$, recovering the quantum regime.

9.2 Coherence length

In a slowly varying substrate the coherence length of a wavefunction is the scale over which Φ varies by $O(1)$:

$$L_{\mathrm{coh}} \;=\; 2\Phi/|\nabla\Phi| \;\sim\; 2r^2/A \quad \text{near a soliton.} \tag{26}$$

This is the TGP analogue of the Compton wavelength: it is finite and position-dependent.

9.3 Quantitative separation of scales

Script `q7_decoherence.py` (8/8 PASS) gives:

- air at STP: $\Gamma_{\mathrm{dec}} \approx 2 \times 10^{28}\ \mathrm{s}^{-1}$ (instant decoherence);
- outer space vacuum: $\Gamma_{\mathrm{dec}} \approx 8 \times 10^{-15}\ \mathrm{s}^{-1}$ (coherence retained over macroscopic times);
- ratio across these environments: 2.4×10^{42}.

The non-linear rate Γ_2 scales with N^2 (verified at machine precision), not N, a prediction distinguishing TGP from standard environment-induced decoherence of Zurek type [6].

9.4 Quantum Darwinism

The geometric picture of decoherence in TGP also delivers Zurek's quantum Darwinism for free:

Theorem 19 (Quantum Darwinism from soliton tails)**.** *For N-soliton environments, the mutual information $I(\mathcal{S} : \mathcal{E})$ saturates at a fraction of the total environment. Numerically, saturation occurs at 23.3% of the environment, giving redundancy $R = 4.3$ (i.e. ~ 4 independent copies of the system's information are imprinted on the substrate).*

The mechanism is geometric: the soliton tail at distance r is shared by every other soliton within $\sim r$, creating $B_{\mathrm{rel}}(r)$ coincident copies automatically. Quantum Darwinism is thus a *geometric* consequence of the tail structure (2), not a separate postulate.

10 Experimental predictions and falsifiability

The structural results of Sections 3–9 yield four testable predictions that would falsify or support TGP.

10.1 Altitude-dependent Planck constant

By Corollary 6, $\hbar(\Phi) = \hbar_0\sqrt{\Phi_0/\Phi}$, so in a gravitational potential U:

$$\frac{\Delta\hbar}{\hbar} \approx -\frac{GM}{rc^2} \approx -3.5 \times 10^{-10} \quad \text{(Earth surface vs. orbit).} \tag{27}$$

Current atom interferometers reach $\sim 10^{-12}$ sensitivity [10, 11], making this shift detectable at the several-σ level in a dedicated drop-tower or space-based interferometer (CAL/ISS class).

10.2 Second-harmonic signature

The amplitude ratio $A_2/A_1 \sim 1\%$ (Sec. 3, Theorem 5) gives a second-harmonic component in elastic scattering at momentum transfer $q = 2A_{\mathrm{tail}}^{-1}$. Present scattering data places an upper bound $A_2/A_1 < 5\%$; a dedicated precision experiment on a low-Z target (electron–proton, $q \sim m_e^{-1}$) could resolve the predicted level.

10.3 Rotational excitation spectrum

Eq. (22) predicts $E(3/2)/E(1/2) = 5$ for the rotational spectrum of any fundamental TGP soliton with $B = 1$. This is the Δ/N ratio of QCD baryons; it also applies to lepton-sector solitons (charged leptons, Sec. 11).

10.4 TGP-specific decoherence scaling

The non-linear channel $\Gamma_2 \sim N^2$ distinguishes TGP from standard environment-induced decoherence rates $\Gamma \sim N$ [6]. In the deep-UV-protected quantum-optics regime (photons or ion traps, where standard N-scaling dominates) the two are indistinguishable. In the dense-medium high-ε regime (condensed-matter substrates, cold atomic clouds approaching the BEC threshold) the quadratic scaling should appear as an excess over linear environment-coupling predictions.

11 Discussion and open problems

We close by collecting four open problems suitable for follow-up work, in order of decreasing specificity.

OP-QM1: Lean 4 formalisation. Theorem 16 (CHSH $= 2\sqrt{2}$) has a sufficiently algebraic structure to admit a machine-checked proof in Lean 4 using the existing Mathlib formalisation of Hilbert spaces. This would turn the numerical verification (machine-precision agreement on a 200×200 grid) into a formal theorem.

OP-QM2: gauge emergence from spin $\frac{1}{2}$. Core-paper problem OP-8 asks for the emergence of the $U(1) \times SU(2) \times SU(3)$ gauge group from the substrate. The spin-$\frac{1}{2}$ result of Sec. 8 (via $\pi_3(S^3) = \mathbb{Z}$) already provides a $SU(2)$ structure on the vielbein. A natural follow-up question: does the full electroweak gauge group emerge from higher components of the vielbein decomposition, and does a colour $SU(3)$ emerge from a three-soliton bound state with the discrete $\mathrm{GL}(3, \mathbb{F}_2)/\mathbb{Z}_3$ structure of [2]?

OP-QM3: second quantisation of Φ. Core-paper problem OP-10 asks for the second quantisation of the substrate field beyond the mean-shift linear regime used throughout this paper. Theorem 12 (the measurement asymmetry) provides a physical handle: a full operator formulation should reproduce the classical asymmetry at tree level and allow loop corrections to $\chi(g_0)$ — in particular, whether the $\sim 7\%$ coefficient of variation of χ over $g_0 \in [0.3, 0.95]$ is an artefact of the classical truncation or a physical prediction for χ running.

OP-QM4: direct detection of $\hbar(\Phi)$. The headline prediction (27) has two experimental signatures: (i) altitude dependence of $\hbar$, testable in atom interferometry at height difference $\Delta h \sim 10$ m with $\sigma(\hbar)/\hbar \sim 10^{-11}$ precision (within reach of current equipment); (ii) a Mach–Zehnder–type test that is sensitive to the *gradient* of $\hbar$ across the interferometer arms rather than to an absolute shift, potentially giving $\sim 10\times$ improved sensitivity.

Closure. Starting from the TGP core paper [1] and its substrate soliton ODE, we have derived the Planck constant, the uncertainty principle, the Born rule, superposition, Bell CHSH $= 2\sqrt{2}$, spin $\frac{1}{2}$, the spin–statistics connection, Fermi–Dirac/Bose–Einstein distributions, and decoherence — that is, the full postulate structure of non-relativistic quantum mechanics — without adding a new degree of freedom. The compiled numerical support (twelve Python scripts totalling 25 tests, with 22 PASS and three non-PASS that reveal the measurement asymmetry) is deposited in the companion repository [4] under `research/qm_foundations/` through `research/qm_decoherence/`.

Only `numpy`, `scipy` and `matplotlib` are required to reproduce all results.

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